Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table? 2

________________show ip route ospf ________________________________

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial0/0/1 ________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __~~65~~______ 129 0

Does R2 show up as an OSPF neighbor on R1? __No______ 0

Does R2 show up as an OSPF neighbor on R3? __No______ 2

What does this information tell you?

____When an interface is set to passive, OSPF stops sending and receiving hello packets on that interface. As a result, no adjacency can form, and the router no longer advertises or learns routes through the passive interface. 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_____R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ____Serial0/0/1 ________ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

________65(serial link cost 64 + Ethernet cost 1)________ 2

Is R2 listed as an OSPF neighbor to R3? __Yes ____ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

_____The default reference bandwidth of 100 Mbps assigns the same cost of 1 to all links of 100 Mbps or faster. This prevents OSPF from correctly distinguishing between Fast Ethernet, Gigabit Ethernet, and higher-speed links. Raising the reference bandwidth enables more precise cost calculations that accurately reflect real interface speeds 2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

____Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

For 192.168.3.0/24: R3 G0/0 cost (1) + R1 S0/0/1 serial cost (64) = total 65. 782 For 192.168.23.0/30: the path is calculated using the sum of the two serial link costs (64 + 64) or the equivalent lowest-cost path between the routers, resulting in 128. 845 0

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

_____845. Because each serial interface bandwidth is now 128 kbps, OSPF recalculates the serial link cost as 781 (100,000,000 / 128,000). Adding the remaining link cost produces the new total of 845. 0

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

______The ip ospf cost 1565 command manually raises the cost of R1's S0/0/1 interface to 1565. This value is higher than the total path cost through R2 (1562), so OSPF selects the lower-cost route via R2 as the best path. ____

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   ____The router ID serves as the unique identifier for each router within the OSPF domain. If two routers accidentally share the same router ID, adjacencies fail to form correctly, LSAs become confused, and routing tables may contain incorrect or unstable information.

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2. Why is the DR/BDR election process not a concern in this lab?

   _____All serial links in this topology are point-to-point networks. OSPF does not perform DR/BDR elections on point-to-point links; the election only occurs on multi-access broadcast or non-broadcast networks.

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3. Why would you want to set an OSPF interface to passive?

   ______Configuring an interface as passive prevents OSPF from sending hello packets and forming unnecessary neighbor relationships. This reduces routing protocol traffic on LAN segments that do not require dynamic routing updates and also improves security by stopping route advertisements on those interfaces.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

_______OSPF divides the network into multiple areas, with Area 0 serving as the backbone. This hierarchical area structure limits the scope of link-state advertisements, reduces SPF calculations, and allows the protocol to support very large networks efficiently.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   ______router ospf 1

   network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   ________The EIGRP route will be installed. OSPF selects routes based on the lowest administrative distance first (EIGRP = 90, OSPF = 110, RIP = 120). Only when AD values are equal does the metric get compared.

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